

STAT 440 Screening Quiz

1. $f(x_1, x_2) = x_1^3 + 3x_1^2x_2 - 3x_1^2 - 3x_2^2 + 2$

a) $\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{pmatrix} \rightarrow \begin{pmatrix} 6x_1x_2 - 6x_1 \\ 3x_1^2 - 6x_2 \end{pmatrix}$ gradient

b) Hessian $H_f(x) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{pmatrix} = \begin{pmatrix} 6x_2 - 6 & 6x_1 \\ 6x_1 & 6x_2 - 6 \end{pmatrix}$

c) Compute values of x_1 and x_2 at which $f(x_1, x_2)$ is minimized
 Set $\nabla f = 0$ for critical points
 $f_{x_1} = 6x_1x_2 - 6x_1 = 0 \quad 6x_1x_2 - 6x_1 = 0 \quad 6x_1x_2 = 6x_1 \quad x_1x_2 = x_1 \quad x_2 = 1 \rightarrow x_1 = 1$
 $f_{x_2} = 3x_1^2 - 6x_2 = 0 \quad 3x_1^2 - 6x_2 = 0 \quad x_1^2 + x_1^2 - 2x_2 = 0 \quad x_1^2 + x_1^2 - 2x_2 = 0$

2. $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 1 & 3 & 4 \end{bmatrix} \quad x = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} \quad y = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$

a) Compute Euclidean norm of x , the norm of x is the length of vector x and length

$\|x\| = \sqrt{2^2 + 1^2 + 5^2} = \sqrt{30}$ unit vector: $\begin{bmatrix} \frac{2}{\sqrt{30}} \\ \frac{1}{\sqrt{30}} \\ \frac{5}{\sqrt{30}} \end{bmatrix}$
 ↳ Euclidean norm

b) Inner product of x and y

$\begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = 2 \cdot 1 + 1 \cdot 3 + 5 \cdot 2 = \boxed{15}$

c) Cosine of angle between x and y

$\cos \theta = \frac{x \cdot y}{\|x\| \|y\|} = \frac{15}{\sqrt{30} \sqrt{14}}$

d) $Ax = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 1 & 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \cdot 2 + 2 \cdot 1 + 3 \cdot 5 \\ 1 \cdot 2 + 4 \cdot 1 + 2 \cdot 5 \\ 1 \cdot 2 + 3 \cdot 1 + 4 \cdot 5 \end{bmatrix} = \begin{bmatrix} 19 \\ 16 \\ 25 \end{bmatrix}$

e) Is A invertible? $\begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 1 & 4 & 2 & | & 0 & 1 & 0 \\ 1 & 3 & 4 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 2 & -1 & | & -1 & 1 & 0 \\ 0 & 1 & 1 & | & -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 2 & -1 & | & -1 & 1 & 0 \\ 0 & 1 & 1 & | & -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 2 & -1 & | & -1 & 1 & 0 \\ 0 & 1 & 1 & | & -1 & 0 & 1 \end{bmatrix}$

$\rightarrow \begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 2 & -1 & | & -1 & 1 & 0 \\ 0 & 1 & 1 & | & -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & -1 & 0 & 1 \\ 0 & 2 & -1 & | & -1 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & -1 & 0 & 1 \\ 0 & 0 & -3 & | & 1 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & | & 1 & 0 & 0 \\ 0 & 1 & 1 & | & -1 & 0 & 1 \\ 0 & 0 & -3 & | & 1 & 1 & -1 \end{bmatrix}$

d) Rank of A?

A can be reduced (terminology) to the identity matrix, which in this case is 3×3 , so it has 3 non-zero rows meaning the rank is $\boxed{3}$

3, $P(A) = 0.5$ $P(B) = P(A \cup B) = 0.7$

• assumes A & B independent, calculate $P(B)$ Emily

• assumes A & B mutually exclusive, calculate $P(B)$ Sam

Find absolute difference between the two calculations

for Sam, $P(B) = 0.2 \rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $0.7 = 0.5 + P(B) - 0$

for Emily, $P(B) = 0.4 \rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $= P(A) + P(B) - P(A)P(B)$

difference $= 0.4 - 0.2 = \boxed{0.2}$
 or 20%

$0.7 = 0.5 + P(B) - 0.5P(B)$

$0.2 = P(B) - 0.5P(B)$

$0.2 = 0.5P(B) \rightarrow P(B) = 0.4$

4, x is a r.v., $Var(x) = E[(x - E(x))^2]$

a) prove that $Var(x) = E(x^2) - E(x)^2$

apply linearity of expectation given the first step: $Var(x) = E[(x - E(x))^2]$

expand $= E(x^2 - 2xE(x) + E(x)^2)$
 $= E(x^2) - E(2xE(x)) + E(E(x)^2)$
 $= E(x^2) - 2E(x)E(x) + E(x)^2$
 $= E(x^2) - E(x)^2$

b) what is the variance of x if $E(x) = 0$ & $E(x^2) = 1$?

$Var(x) = E(x^2) - E(x)^2 = 1 - 0^2 = \boxed{1}$

c) assume $y = a + bx$, where a & b are constants, what is the variance of y ?

$Var(y) = Var(a + bx) = Var(a) + b^2 Var(x) = b^2 Var(x)$, so $Var(y) = b^2$

As constant has no variance, there is no

variance term between a constant & a r.v.

5. assume that the value of medical claims x , made in one year at an insurance

company is described by

$x \sim U(0, 1)$, where x is a r.v. w/ pdf $f(x) = \begin{cases} c(1-x)^4, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$

where c is constant

a) find cdf of x $\int_0^x c(1-t)^4 dt = c \int_0^x (1-t)^4 dt = \left[-\frac{c(1-t)^5}{5} \right]_0^x = \frac{c(1-x)^5}{5} - \frac{c}{5}$

$$\begin{aligned}
 5) E(x) &= \int_0^1 x \cdot c(1-x)^4 dx = c \int_0^1 x(1-x)^4 dx \quad u=1-x \text{ so } x=1-u \\
 &= c \int_0^1 (1-u)u^4 du \quad \text{du} = -dx \\
 &= c \int_0^1 (u^4 - u^5) du = c \left[\frac{u^5}{5} - \frac{u^6}{6} \right]_0^1 = c \left(\frac{1}{5} - \frac{1}{6} \right) = \boxed{c/30}
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(x) &= E(x^2) - E(x)^2 \quad E(x)^2 = c/900 \\
 E(x^2) &= c \int_0^1 x^2(1-x)^4 dx \quad u=1-x \text{ so } x=1-u \\
 &= c \int_0^1 (1-2u+u^2)(u^4) du = c \int_0^1 (u^4 - 2u^5 + u^6) du \\
 &= c \left[\frac{u^5}{5} - \frac{2u^6}{6} + \frac{u^7}{7} \right]_0^1 = c \left(\frac{1}{5} - \frac{1}{3} + \frac{1}{7} \right) = \frac{c}{105} \\
 \text{Var}(x) &= \frac{c}{105} - \frac{c}{900}
 \end{aligned}$$

c) Calculate $P(V > 10,000)$
 $1 - P(V < 10,000)$ plug in the cdf $V(10,000)$
 for V to be $< 10,000$, x must be < 10
 $1 - P(x < 10)$
 $= 1 - \frac{(10+1)^5}{5} = 1 - \frac{(11)^5}{5} = \frac{1 - 161051}{5}$ alright then

6). Bowl 1 Bowl 2
 RR 33 RR RR
 BB 33 BB 33
 to confuse why this wouldn't just be 1/2?
 Bowl 2 could be 4R 3 2B or 3R 3 3B after the transfer, both have 1/2 probability
 Ball picked from Bowl 1, then transferred to Bowl 2, then ball chosen from Bowl 2. Ball chosen from 2 was red, find probability that ball transferred from 1 to 2 was red.

$$\begin{aligned}
 7). P(x=0, y=-1) &= \frac{1}{9} \quad P(x=y=1) = \frac{1}{9} + \frac{1}{6} = \frac{5}{18} \quad P(x=0) = \frac{2}{9} \quad P(x=y) = \frac{1}{18} + \frac{1}{9} + \frac{1}{9} = \frac{5}{18} \\
 8) P(x=-1 | y=1) &= P(x=-1, y=1) / P(y=1) \\
 &= \frac{1}{18} / \left(\frac{1}{18} + \frac{1}{9} + \frac{1}{6} \right) = \frac{1}{18} / \frac{5}{18} = \frac{1}{5}
 \end{aligned}$$

8. 9 questions, 5 answer choices (one is correct)

2 people answer all randomly. Find probability they get same # of answers correct
Binomial probability, $X = \text{correct answers}$

$p = 0.2$
 $n = 9$

$$P(X=0) = \binom{9}{0} (0.2)^0 (0.8)^9 = 0.32768$$

$$P(X=1) = \binom{9}{1} (0.2)^1 (0.8)^8 = 0.4096$$

$$P(X=2) = \binom{9}{2} (0.2)^2 (0.8)^7 = 0.2048$$

$$P(X=3) = \binom{9}{3} (0.2)^3 (0.8)^6 = 0.0912$$

$$P(X=4) = \binom{9}{4} (0.2)^4 (0.8)^5 = 0.0064$$

$$P(X=5) = \binom{9}{5} (0.2)^5 (0.8)^4 = 0.00032$$

Independent events, so $P(A \cap B) = P(A)P(B)$ where $A = \text{student's score}$
 $B = \text{the other student's score}$

Mutually exclusive events, neither student could have gotten 9 answers correct AND 3 answers correct, so

$$P(\text{both 0 right} \cup 1 \text{ right} \cup 2 \text{ right} \dots \cup 5 \text{ right}) \\ = P(\text{both 0 right}) + P(\text{both 1 right}) + \dots + P(\text{both 5 right})$$

$$\text{Final probability: } 0.32768^2 + 0.4096^2 + 0.2048^2 + 0.0912^2 + 0.0064^2 + 0.00032^2 \\ \approx 0.10737 + 0.16777 + 0.04194 + 0.00262 + 0.00004096 + 0.0000001024$$

$$\approx \boxed{0.3197}$$

Look for a normal d.v. X , $P(X < 500) = 0.5$ and $P(X > 650) = 0.0227$, find standard deviation of X

$$\mu = 500$$

$$P(Z > 500) > 650 = 0.0227$$

$$P(Z > 150) = 0.0227$$

$$P\left(Z > \frac{150}{\sigma}\right) = 0.0227$$

$$0.5 - P\left(0 < Z < \frac{150}{\sigma}\right) = 0.0227$$

$$P\left(0 < Z < \frac{150}{\sigma}\right) = 0.4773$$

$$P(0 < Z < 2) = 0.4773$$

$$\frac{150}{\sigma} = 2 \text{ so } \boxed{\sigma = 75}$$

10. $S = \{7, 6, 8, 9, 2, 7, 6, 7, 6, 5\}$ $n = 10$

a) Sample mean:

$$\sum x_i = 7+6+8+9+2+7+6+7+6+5 = 58$$

$$\bar{x} = 58/10 = \boxed{5.8}$$

b) Sample variance: $\frac{\sum_{i=1}^{10} (x_i - \bar{x})^2}{10-1}$

$$= \frac{(7-5.8)^2 + (6-5.8)^2 + (8-5.8)^2 + (9-5.8)^2 + (2-5.8)^2 + (7-5.8)^2 + (6-5.8)^2 + (7-5.8)^2 + (6-5.8)^2 + (5-5.8)^2}{9}$$

$$= \boxed{3.06}$$